OriPics GPSA v2.2.2

OriPics — Generator Product Security Architecture Document

> **Applicant**: SantaHades Co., Ltd. **Record ID**: 019e4988-9d7c-72f8-8675-22eacf3e1904 **Version**: 2.2.2 (2026-05-30) **Supersedes**: v2.2.1 (2026-05-29), v2.2 (2026-05-29), v2.1 (2026-05-28), v2.0 (2026-05-24), v1.0 (2026-05-22) **Target Assurance Level**: Level 1

Summary of Changes in v2.2.2

v2.2.2 remediates the two nonconformities identified in the C2PA conformance review of v2.2.1 (Scott S. Perry, 2026-05-30):

1. Validation logic now consults the C2PA Trust List. The OriPics verification function (used by the public verify feature) previously did not load the C2PA Trust List when running in production mode, so even a manifest signed by a certificate chaining to the Trust List would be reported as `untrusted`. The verifier now loads the official **C2PA Conformance Trust List** — `C2PA-TRUST-LIST.pem` (signing CAs) and `C2PA-TSA-TRUST-LIST.pem` (timestamp CAs) from `c2pa-org/conformance-public`, plus the EKU trust configuration — via `createTrustSettings()`, with `verifyTrust` and `verifyTimestampTrust` enabled. A signing certificate chaining to the Trust List (e.g. `SSL.com C2PA ECC/RSA Root CA 2025`) now yields a `trusted` verdict.

2. Standard-tier (ingested file) action corrected to `c2pa.opened`. OriPics does not create the asset for a Web upload / file pick — the user supplies it — so the Generator Product has no right to assert `c2pa.created` (which claims creation of the asset itself, not merely of the manifest). The only conformant inception action for an ingested file is **`c2pa.opened`** (no `digitalSourceType`), with the uploaded file recorded as a **`parentOf` ingredient of unknown provenance** (OriPics did not create it and cannot prove C2PA creation provenance). This supersedes the v2.2.1 `c2pa.created` decision.

Scenario	Action (v2.2.2, as implemented and as in the attached samples)
Standard tier (Web upload / mobile file pick), no prior manifest	**`c2pa.opened`** (no `digitalSourceType`) + uploaded file as `parentOf` ingredient of **unknown provenance**
Verified tier (mobile native camera, App Attest / Play Integrity)	`c2pa.created` + `digitalSourceType: digitalCapture`
Any tier, prior C2PA manifest present	`c2pa.opened` (no `digitalSourceType`) + prior manifest as `parentOf` ingredient

Rationale: `c2pa.created` asserts creation of the asset; OriPics may legitimately assert it only for Verified-tier captures, where in-app camera hardware (confirmed by App Attest / Play Integrity) actually produces the asset within the TOE. For all ingested files, OriPics asserts only that it opened an existing file (`c2pa.opened`) and seals it; if creation attribution were ever required, the CAWG identity assertion (in `gathered_assertions`) would be the conformant mechanism — OriPics does not claim creation and therefore does not use it.

Summary of Changes in v2.2.1

This is an editorial consistency correction of v2.2. **The implemented behavior is unchanged** — the v2.2 *Summary of Changes* already specified `c2pa.created` with no `digitalSourceType` for Standard-tier uploads, and the implementation and sample files have always produced exactly this. However, several v2.2 body sections inconsistently described the Standard-tier action as `c2pa.published`, `c2pa.opened`, or `c2pa.created` + `digitalCapture`. v2.2.1 corrects all body sections, tables, the architecture diagram, and the attachment description to consistently state the actual, conformant behavior:

Scenario	Action (v2.2.1, as implemented and as in the attached samples)
Standard tier (Web upload / mobile file pick), no prior manifest	`c2pa.created` with **no `digitalSourceType`** (no capture claim)
Verified tier (mobile native camera, App Attest / Play Integrity)	`c2pa.created` + `digitalSourceType: digitalCapture`
Any tier, prior C2PA manifest present	`c2pa.opened` + `c2pa.ingredient.v3` (`relationship: "parentOf"`)

Rationale: per the C2PA specification, the first action of a parentless manifest must be `c2pa.created` or `c2pa.opened` (a standalone `c2pa.published` first action fails validation with `assertion.action.malformed`). For unverified uploads, OriPics therefore uses `c2pa.created` **without** any `digitalSourceType` — asserting only that OriPics created the C2PA manifest, and making **no claim that the underlying image was captured or otherwise originated by OriPics**. The `digitalCapture` source type is reserved exclusively for Verified-tier mobile captures confirmed by device attestation.

Summary of Changes in v2.2

Item	v2.1	v2.2
`c2pa.actions.v2` assertion placement	`gathered_assertions` (via `addAssertion()`) — nonconformant	**`created_assertions`** (via `builder.addAction()` per action) — conformant
Web client in TOE boundary	Included as Edge subsystem	**Explicitly excluded from TOE** — Web is content-submission portal only; Req 6.2.1.5/6 satisfied by mobile clients (App Attest/Play Integrity)
Action for Standard tier (web/mobile file pick)	`c2pa.created` + `digitalCapture` — incorrect (capture not verifiable)	**`c2pa.created`** (no `digitalSourceType`); only Verified tier (mobile camera, App Attest/Play Integrity) uses `c2pa.created` + `digitalCapture`
Signing key storage language	"SSL.com eSigner Cloud HSM (production)" — tentative/unimplemented	**Clearly stated**: current production = Vercel encrypted env vars (AES-256); HSM is planned Phase 2 post-conformance
TOE for Req 6.2.1.5/6	Web + Mobile	**Mobile only** (iOS App Attest + Android Play Integrity); Web explicitly excluded

Summary of Changes in v2.1

Item	v2.0	v2.1
TLS for Backend→Supabase	TLS 1.2+	**TLS 1.3** enforced via `instrumentation.ts` `tls.DEFAULT_MIN_VERSION`
Pre-existing manifest handling (§C.2.7)	"Pending guidance" (not implemented)	**Approach A implemented**: `c2pa.opened` + `parentOf` ingredient
Action for ingested file	`c2pa.created` (incorrect)	**`c2pa.opened`** when prior manifest detected
Assertion classification note	"All in created_assertions" (incorrect)	Corrected: c2pa-rs Claim v2 places hash in created, data in gathered
Attachments	Standard sample only	**+ingredient sample** (Pixel Camera JPG from Interoperability Library)

Summary of Changes in v2.0

Item	v1.0	v2.0
Implementation Class	Backend	**Distributed**
TOE Architecture	Backend-only diagram	Distributed diagram (3 Edge clients + Backend)
Edge-to-Backend mutual auth	Not documented	Documented (§C.2.2 §4)
Digital Source Type	`algorithmicMedia` (incorrect)	**`digitalCapture`**
Capture-only / Ingredient policy	Not documented	New §C.2.7

C.1. Generator Product Information

C.1.1. Applicant Organization Details

- **Legal Name**: SantaHades Co., Ltd. (주식회사 산타하데스)
- **Address**: #B01-H306, Terrace Garden, 150-29 Gongse-ro, Giheung-gu, Yongin-si, Gyeonggi-do 17084, Republic of Korea
- **Contact Person**: Yongseog Son, Representative Director and CEO
- **Email**: hi@ori.pics
- **Phone**: +82-10-3717-9717
- **Security Contact**: security@ori.pics (RFC 9116 security.txt published at https://www.ori.pics/.well-known/security.txt)

C.1.2. Distinguished Name

Field	Value
Common Name (CN)	OriPics
Organization (O)	SantaHades Co., Ltd.
Organizational Unit (OU)	*(not applicable — single-team organization)*
Country (C)	KR

C.1.3. Generator Product Description

OriPics is a photo provenance and authenticity platform for the Korean consumer and SMB market, with planned expansion to global users. It allows users to prove that an image is original (untampered) and to certify when and where it was captured.

Platform coverage and TOE classification:

Platform	TOE membership	Role
OriPics iOS	**Within TOE**	Native camera capture (Verified tier, App Attest) + file pick (Standard tier). Device integrity: Apple App Attest.
OriPics Android	**Within TOE**	Native camera capture (Verified tier, Play Integrity) + file pick (Standard tier). Device integrity: Google Play Integrity.
OriPics Web	**Outside TOE**	Browser-based content submission portal only. Browser JS cannot provide cryptographic subsystem authentication. Web uploads are ingested via a `c2pa.opened` action (no `digitalSourceType`) with the uploaded file recorded as a `parentOf` ingredient of **unknown provenance** — OriPics did not create the asset. Not a conformant Edge subsystem.

> **TOE boundary justification**: The C2PA Generator Product TOE requires that Edge subsystems be mutually authenticated with the Backend (Req 6.2.1.5/6). The mobile clients satisfy this requirement via Apple App Attest (iOS) and Google Play Integrity (Android). The Web client runs in a browser JavaScript environment that provides no secure storage or execution boundary; it cannot be cryptographically authenticated as a subsystem instance. Accordingly, the Web client is **explicitly excluded from the TOE** for C2PA manifest generation purposes. Web uploads are accepted as user-submitted content; the Backend ingests them with a `c2pa.opened` action and records the uploaded file as a `parentOf` ingredient of **unknown provenance** — asserting only that OriPics opened and sealed an existing file, and making no claim that OriPics created, captured, or originated the underlying image.

The mobile and backend clients share a common backend signing service hosted on Vercel Functions. The backend holds the C2PA signing certificate; no signing key material is present on any edge device.

> **Note on separate submissions**: Per C2PA Conformance Program requirements, OriPics iOS and OriPics Android will be submitted as separate generator products with separate Intake Forms. This document covers the full Distributed architecture shared by all platforms. The backend signing service and certificate are common to all platforms; platform-specific details are called out where relevant.

Intended use cases:

- Traffic accident scene documentation for insurance and legal submissions
- Real estate move-in/move-out condition evidence
- Creator portfolio originality proof
- Media and journalism photo provenance
- Dating and social media identity verification

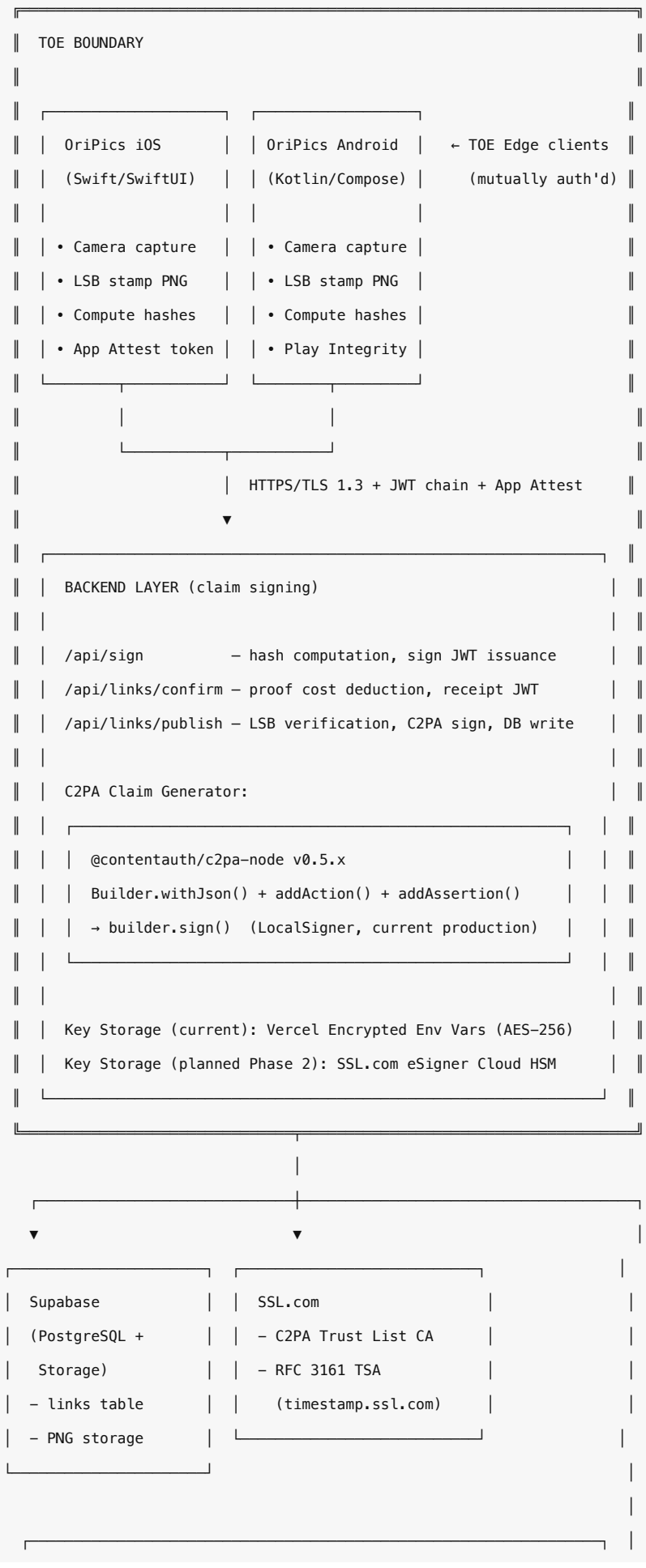
Key features:

- Steganographic pixel-integrity stamping (proprietary LSB embedding + HMAC verification)
- C2PA manifest generation and attachment to output PNG images
- Two trust tiers: Standard (web upload or mobile file pick) and Verified (mobile native camera capture with device attestation)
- Preservation of provenance chain: pre-existing C2PA manifests are included as `c2pa.ingredient` assertions
- Public shareable links with 7-day (Free) or permanent (Pro/Business) retention

Target audience: Individual consumers (B2C, traffic accidents, personal proof) and small businesses (B2B, real estate agencies, media organizations).

C.1.4. Generator Product Target of Evaluation (TOE) Description

The OriPics Generator Product TOE uses a **Distributed Implementation Class**. Claim preparation happens on Edge devices; claim signing happens on the Backend service.


TOE BOUNDARY
OriPics iOS
(Swift/SwiftUI)
• Camera capture
• LSB stamp PNG
• Compute hashes
• App Attest token
OriPics Android
(Kotlin/Compose)
• Camera capture
• LSB stamp PNG
• Compute hashes
• Play Integrity
← TOE Edge clients
(mutually auth'd)
HTTPS/TLS 1.3 + JWT chain + App Attest
BACKEND LAYER (claim signing)
/api/sign — hash computation, sign JWT issuance
/api/links/confirm — proof cost deduction, receipt JWT
/api/links/publish — LSB verification, C2PA sign, DB write
C2PA Claim Generator:
@contentauth/c2pa-node v0.5.x
Builder.withJson() + addAction() + addAssertion()
→ builder.sign() (LocalSigner, current production)
Key Storage (current): Vercel Encrypted Env Vars (AES-256)
Key Storage (planned Phase 2): SSL.com eSigner Cloud HSM
Supabase
(PostgreSQL +
Storage)
- links table
- PNG storage
SSL.com
- C2PA Trust List CA
- RFC 3161 TSA
(timestamp.ssl.com)

```
  │  OriPics Web (browser JS)   ← OUTSIDE TOE                       │  │
  │                                                                 │  │
  │  • LSB stamp PNG (client-side)                                  │  │
  │  • Compute hashes                                               │  │
  │  • No cryptographic subsystem authentication possible           │  │
  │  • Backend issues c2pa.opened for uploads (unknown-prov. ingr.) │  │
  └─────────────────────────────────┬───────────────────────────────┘  │
                                    │ HTTPS/TLS 1.3                    │
                                    └──────────────────────────────────┘
```

Platform runtime:

- Backend: Vercel Serverless Functions (AWS-backed), Node.js 18+ runtime
- Edge (Web): Browser JavaScript (Chrome, Safari, Firefox — all modern browsers)
- Edge (iOS): Swift/SwiftUI, iOS 16+, ARM64
- Edge (Android): Kotlin/Jetpack Compose, Android 11+, ARM64

Third-party services in TOE scope:

Service	Role	Security Properties
Vercel	Hosting, serverless compute, CDN, TLS termination	SOC 2 Type II, encrypted env vars, immutable deployments
Supabase	PostgreSQL database, object storage (S3-compatible)	SOC 2 Type II, RLS policies, AES-256 encryption at rest
SSL.com	C2PA Trust List CA, eSigner Cloud HSM, TSA	C2PA Trust List registered CA (`SSL.com C2PA ECC/RSA Root CA 2025`), WebTrust audited
GitHub	Source code repository, CI/CD	Branch protection, Dependabot, CodeQL, secret scanning
Apple App Attest	iOS device integrity attestation (Verified tier)	Apple-managed PKI, attestation key bound to device Secure Enclave
Google Play Integrity	Android device integrity attestation (Verified tier)	Google-managed, Play Integrity API

C.1.5. Implementation Class

Distributed

The OriPics Generator Product is classified as **Distributed** because claim generation spans both Edge and Backend subsystems:

Subsystem	TOE	Role in Claim Generation
Edge — OriPics iOS	**In TOE**	Native camera capture; steganographic LSB stamping; HMAC hash computation; App Attest token for Verified tier; PNG upload via signed URL
Edge — OriPics Android	**In TOE**	Same as iOS; Google Play Integrity for Verified tier
Edge — OriPics Web	**Outside TOE**	Content submission portal only. LSB stamping and hash computation performed client-side, but no cryptographic subsystem authentication possible. Backend ingests all Web uploads with `c2pa.opened` (no `digitalSourceType`) + uploaded file as `parentOf` ingredient of unknown provenance.
Backend — Vercel Functions	**In TOE**	Receives hashes from Edge via HTTPS; verifies mobile Edge identity via JWT chain + LSB hash extraction + App Attest/Play Integrity; constructs C2PA manifest; signs manifest using `@contentauth/c2pa-node`; writes to Supabase

Signing key location: The C2PA signing private key is held exclusively by the Backend. **Current production**: Vercel Encrypted Environment Variables (AES-256). **Planned Phase 2**: SSL.com eSigner Cloud HSM via CSC API (not yet implemented as of v2.2). No signing key material is present on any Edge device.

Claim generator values:

- Standard tier (all platforms): `oripics/0.1.0`
- Verified tier (mobile native capture): `oripics/0.1.0 (mobile-native)`

C.1.6. Target Max Assurance Level

Level 1

C.1.7. Target Generator Product Capabilities

Claim generation:

- Still image media types:
  - image/png

Claim validation: OriPics is submitted as a **Generator Product**. It additionally operates an internal verification function (`readC2paManifest()`, backing the user-facing verify feature) that reads a manifest and reports its trust status. This verifier **consults the C2PA Trust List** (see §C.2.4 §5): it loads the official C2PA Conformance Trust List (signing + TSA CAs) and reports `trusted` only when the signing certificate chains to it within validity, as established by the trusted timestamp. Authoritative third-party validation remains available via tools such as contentcredentials.org/verify.

C.2. Security Architecture Details

C.2.1. Authentication for Certificate Enrollment

Certificate Enrollment Process

OriPics uses SSL.com as its Certification Authority (CA), which is on the C2PA Trust List.

Current (pre-production / sandbox):

1. CSR generated with OpenSSL (ECC P-256)
2. CSR submitted to SSL.com c2pasign.com sandbox for test cert issuance
3. Test cert + private key stored as Vercel encrypted environment variables
4. LocalSigner in `@contentauth/c2pa-node` uses the cert+key to sign manifests

Production (post-Conformance Letter):

1. SSL.com issues a C2PA Claim Signing Certificate to OriPics after identity verification (OV-level KYC)
2. Private key is generated and stored within SSL.com eSigner Cloud HSM — the private key never leaves the HSM
3. OriPics backend authenticates to SSL.com eSigner CSC API using OAuth 2.0 client credentials (client_id + client_secret)
4. Each signing operation is a remote API call: the backend sends the hash to be signed, SSL.com HSM signs it, and returns the signature
5. Certificate rotation: SSL.com manages certificate lifecycle; OriPics updates Vercel environment variables when a new cert is issued

Management of Certificate Enrollment Authentication Secrets

(Required for Assurance Level 1)

Authentication method: OAuth 2.0 client credentials (client_id + client_secret) for SSL.com eSigner CSC API.

Secret management:

- `ORIPICS_ESIGNER_CLIENT_ID` and `ORIPICS_ESIGNER_CLIENT_SECRET` are stored in Vercel encrypted environment variables
- Vercel encrypts environment variables at rest using AES-256 and restricts access via team RBAC
- Environment variables are scoped to Production environment only (Preview/Development use separate sandbox credentials)
- Secrets are never committed to source code; `.env` files are gitignored
- GitHub secret scanning is enabled on the repository to detect accidental credential exposure
- Access to Vercel project settings (where env vars are managed) requires team owner authentication with MFA enabled

C.2.2. Key Generation, Storage, and Usage

Key Generation and Storage Method

(Required for Assurance Level 1)

1. Key generation and storage:

- **Algorithm**: ECDSA with P-256 (ES256), as required by C2PA specification
- **Current production**: Key generated locally via OpenSSL, stored as PEM in Vercel encrypted environment variable (`ORIPICS_C2PA_KEY_PEM`). Vercel encrypts environment variables at rest with AES-256. The key is decrypted at function cold start, held in Lambda memory only for the duration of the signing operation, and not persisted to disk. This satisfies Level 1 Req 6.2.1.1 (encrypted storage).
- **Planned Phase 2 (not yet implemented)**: Migrate to SSL.com eSigner Cloud HSM (FIPS 140-2 Level 3). Private key would be generated within the HSM and never leave it. OriPics code would call the CSC API for signing operations. This enhancement is planned post-conformance and is not evaluated in this submission.

2. Key access controls:

- Only the Vercel Functions runtime in the Production environment can access the signing credentials
- Vercel environment variables are encrypted at rest (AES-256) and in transit (TLS 1.3)
- Access to the key PEM in Vercel is restricted to the single Vercel team owner (MFA-protected); no other team members exist
- Application code accesses the key only within the `/api/links/publish` route handler; no other code path reads the signing key
- No signing key material is present on any Edge device (Web browser, iOS, Android)

3. Key rotation process:

- **Sandbox**: Manual rotation by generating a new key pair and updating Vercel env vars
- **Production (SSL.com HSM)**: Certificate lifecycle managed by SSL.com. Upon certificate expiration or revocation, OriPics updates the credential ID in Vercel env vars. The old key is destroyed within the HSM by SSL.com per their key management policy
- Key rotation can be performed with zero downtime by deploying updated env vars to Vercel (atomic deployment model)

4. Edge-to-Backend mutual authentication (Distributed Implementation Class):

> **TOE scope**: Req 6.2.1.5/6 (mutual authentication of Edge and Backend subsystems) applies **only to the mobile Edge clients (iOS, Android)** which are within the TOE. The Web client is explicitly outside the TOE and is not subject to subsystem mutual authentication requirements. The Backend accepts Web uploads but ingests them with a `c2pa.opened` action (no `digitalSourceType` ; uploaded file recorded as a `parentOf` ingredient of unknown provenance) — making no creation or capture claim, consistent with its content-submission-only role.

The OriPics Distributed architecture uses a three-stage JWT chain to authenticate the **mobile** Edge client to the Backend signing service. This mechanism verifies that: (a) the user is authenticated (NextAuth session); and (b) the PNG uploaded to Storage was produced by a legitimate OriPics Edge client (LSB hash verification).

Stage 1 — /api/sign (Backend):

1. Validates NextAuth session; checks credit balance
2. Computes `final_hash = HMAC-SHA256(salt, meta_bytes ‖ inner_hash ‖ border_hash)` where `inner_hash` and `border_hash` are computed by the Edge from the original image pixels

3. Issues a sign JWT (`aud: links/confirm`, TTL 5 minutes) containing `{ user_id, tier, link_id, storage_path, timestamp, width, height, final_hash_hex }`
4. Returns `final_hash` (hex) and a Supabase signed upload URL to the Edge

Stage 2 — Edge (all clients):

1. Embeds `meta_bytes ‖ final_hash` into the PNG border pixels via LSB steganography
2. Uploads the stamped PNG to Supabase Storage using the signed upload URL
3. Calls `/api/links/confirm` with the sign JWT

Stage 3 — /api/links/confirm (Backend):

1. Verifies sign JWT signature, expiry, and audience
2. Deducts proof credits
3. Issues a receipt JWT (`aud: links/publish`, TTL 30 days) containing all sign JWT claims including `final_hash_hex`

Stage 4 — /api/links/publish (Backend):

1. Verifies receipt JWT signature, expiry, and audience
2. Verifies `claims.user_id === sessionUserId` (prevents cross-user replay)
3. Downloads the stamped PNG from Supabase Storage
4. **LSB hash extraction and verification**: Decodes PNG pixels using `pngjs`, reads border-pixel LSBs using `extractPayloadV4()`, extracts embedded `final_hash` from the payload. Compares against `final_hash_hex` from the receipt JWT using `crypto.timingSafeEqual()`. Mismatch → 422 response + credit refund
5. Only if hash matches: constructs and signs the C2PA manifest

This mechanism ensures that a tampered PNG (one where pixel content was swapped after the sign JWT was issued but before publish) will fail LSB verification and be rejected before C2PA signing.

Backend-to-SSL.com authentication:

- OriPics Backend authenticates to SSL.com eSigner CSC API using OAuth 2.0 client credentials
- SSL.com authenticates the backend by validating the client credentials against the registered OriPics account
- All communication between the backend and SSL.com is over HTTPS/TLS 1.3

Mobile Edge device attestation (Verified tier):

- iOS: Edge client obtains an Apple App Attest attestation token (`DCAppAttestService`), bound to the device Secure Enclave key and a server-issued nonce. Backend verifies the attestation chain against Apple's root CA
- Android: Edge client obtains a Google Play Integrity verdict token, bound to the app's Play Store identity and a server-issued nonce. Backend verifies the token against Google's Play Integrity API
- Attestation token hash is stored in the `com.oripics.verified` C2PA assertion for audit

C.2.3. Protections Against Claim Generator Misconfiguration and Abuse

Processes for Detecting Vulnerabilities in Dependencies (SCA/SBOM)

(Required for Assurance Level 1)

1. SCA/SBOM dependency vulnerability scanning tools:

- **GitHub Dependabot**: Enabled on the repository (`SantaHades/OriPics-MVP-Front`). Monitors all npm dependencies for known CVEs from the GitHub Advisory Database (NIST NVD data). Configured via `.github/dependabot.yml`:
  - Major version updates: blocked (manual review required)
  - Minor/patch updates: grouped weekly PRs
  - Security updates: automatic PRs for CRITICAL and HIGH severity
- **GitHub CodeQL**: Enabled for JavaScript/TypeScript static analysis. Runs on every PR and weekly scheduled scans. Detects OWASP Top 10 vulnerabilities (SQL injection, XSS, command injection, path traversal, etc.)
- **GitHub Secret Scanning**: Enabled to detect accidental credential commits (API keys, certificates, tokens)
- **npm audit**: Run as part of the development workflow; `pnpm audit` checks against the npm advisory database

2. Build and deployment pipeline vulnerability prevention:

- Dependabot automatically creates PRs for CRITICAL and HIGH severity vulnerabilities
- The development team triages vulnerability reports within 7 days of detection
- CRITICAL and HIGH vulnerabilities are fixed or mitigated within 90 days (C2PA Security Requirements §6.3.1)
- Vercel deployments are immutable: each deployment is a frozen snapshot of the code at a specific Git commit. Once deployed, the running code cannot be modified
- GitHub branch protection on `main` requires PR review and CI passage before merge
- Historical evidence: 21 Dependabot advisories triaged on 2026-05-09; Next.js 14.1.0 → 14.2.35 security patch applied (16 advisories resolved); `next-intl` prototype pollution fix applied 2026-05-10

C.2.4. Protections Against Misconfiguration and Abuse of Software That Processes or Modifies Digital Content or Assertions

(Required for Assurance Level 1)

The software that processes and/or modifies Digital Content in the OriPics TOE includes:

1. **Client-side steganographic stamping** (`oripics-stamp` library, runs on Edge): Embeds integrity data into image border pixels via LSB encoding. Hash computation uses `crypto.subtle.digest()` (Web Crypto API on browser/mobile)
2. **Server-side C2PA manifest construction and signing** (`@contentauth/c2pa-node`, runs on Backend): Constructs and signs the C2PA JUMBF manifest box appended to the output PNG

1. Action selection and Digital Source Type (DST):

Action selection is determined by two factors: (a) whether a prior C2PA manifest exists in the input asset, and (b) whether the claim was generated from a verified mobile camera capture (Verified tier, within TOE) or a web/file-pick upload (Standard tier, outside or peripheral to TOE):

Scenario	Primary action	`digitalSourceType`	Ingredient	TOE
Verified tier, no prior manifest (mobile camera, App Attest/Play Integrity confirmed)	`c2pa.created`	`digitalCapture`	None	Within TOE
Standard tier, no prior manifest (web upload or mobile file pick)	**`c2pa.opened`**	None	uploaded file, `parentOf`, **unknown provenance**	Web = outside TOE
Any tier, prior C2PA manifest present	`c2pa.opened`	None	prior manifest, `parentOf`	—

`c2pa.created` asserts creation of the **asset itself** and is therefore used **only** for Verified-tier captures, where in-app camera hardware (confirmed via App Attest or Play Integrity) directly produced the asset within the TOE; in that case `digitalSourceType: digitalCapture` is also asserted. For all ingested files (Standard tier and any upload carrying a prior manifest), OriPics did not create the asset and uses **`c2pa.opened`** instead.

> **C2PA spec / conformance note** (per Scott S. Perry, 2026-05-30): A Generator Product must not assert `c2pa.created` for an asset it did not create — `created` is a claim about creation of the asset, not merely of the manifest. The only conformant inception action for an ingested file is **`c2pa.opened`**, with the ingested file recorded as a `parentOf` ingredient of **unknown provenance** (OriPics did not create it and cannot prove C2PA creation provenance). `c2pa.opened` requires that ingredient; the SDK adds it automatically via `builder.setIntent('edit')` after `builder.addIngredient(... relationship: "parentOf")`. (A standalone `c2pa.published` first action is likewise nonconformant for a parentless manifest, failing with `assertion.action.malformed`.) If creation attribution were ever required, the conformant mechanism would be a CAWG identity assertion in `gathered_assertions`; OriPics makes no creation claim and does not use it.

Implementation: `apps/web/src/lib/oripics-stamp/c2pa.ts` (`attachC2paManifest()`).

2. Assertions and Claim v2 structure:

OriPics uses `@contentauth/c2pa-node v0.5.x` (backed by `c2pa-rs v0.82.1`) with Claim v2.

Actions are placed in `created_assertions` — the conformant location per C2PA Claim v2 spec. For the `c2pa.created` case (Verified capture), each action is added via `builder.addAction(JSON.stringify(action))` (the `builderAddAction` Neon binding). For the `c2pa.opened` case (Standard upload / prior manifest), the inception action is generated by the SDK from `builder.setIntent('edit')` after `builder.addIngredient(... relationship: "parentOf")`, which lets c2pa-rs compute the ingredient hash reference at signing time; the platform action (`com.oripics.*`) is then appended via `addAction()`. In all cases `c2pa.actions.v2` lands in `created_assertions`. (Using `builder.addAssertion('c2pa.actions.v2', ...)` would place it in `gathered_assertions`, which is nonconformant; this approach is not used.)

Custom assertions (`com.oripics.*`) are added via `builder.addAssertion()` and are placed in `gathered_assertions`. This is correct for non-standard vendor assertions.

Assertion label	Claim v2 placement	How added	Content
`c2pa.hash.data`	`created_assertions`	Auto-generated by c2pa-rs	Cryptographic hash binding
`c2pa.actions.v2`	**`created_assertions`**	`builder.addAction()` / `setIntent('edit')`	`c2pa.created` (Verified capture) or `c2pa.opened` (Standard upload / prior manifest) + platform action
`com.oripics.proof`	`gathered_assertions`	`builder.addAssertion()`	Tier, link_id, verify_url, stamp_version, dimensions, optional GPS
`com.oripics.verified`	`gathered_assertions`	`builder.addAssertion()`	Platform, attest_token_hash, device_integrity *(Verified tier only)*

3. SCA/SBOM scanning for all content-processing software:

- Same tools as §C.2.3: GitHub Dependabot + CodeQL + npm audit cover all npm dependencies including `@contentauth/c2pa-node` and `pngjs`
- `@contentauth/c2pa-node` is the official C2PA SDK maintained by the Content Authenticity Initiative (CAI). Security updates are tracked via Dependabot
- `pngjs` (used for server-side LSB extraction) is monitored by Dependabot
- Vercel's immutable deployment model ensures that a specific deployment always runs the exact dependency versions present at build time

4. Vulnerability remediation pipeline:

- Same 90-day CRITICAL/HIGH remediation policy as §C.2.3
- The `@contentauth/c2pa-node` package includes native binary addons (Rust-compiled); Dependabot monitors its npm advisory entries

5. C2PA Trust List consumption in the verification function (validation logic):

The OriPics verification function (`readC2paManifest()` in `apps/web/src/lib/oripics-stamp/c2pa.ts` , backing the public verify feature and the `/api/verify` and `/api/links/[id]/c2pa` routes) loads the **official C2PA Trust List** when validating a manifest's signature, so that a certificate chaining to the Trust List yields a `trusted` verdict:

- **Trust anchors**: the C2PA Conformance Trust List — `C2PA-TRUST-LIST.pem` (signing CAs) combined with `C2PA-TSA-TRUST-LIST.pem` (timestamp CAs) from the `c2pa-org/conformance-public` repository — is bundled in the product (`apps/web/src/lib/oripics-stamp/c2pa-trust-list.ts` ; refreshed via `scripts/update-c2pa-trust-list.mjs`) and supplied to `createTrustSettings({ verifyTrustList: true, trustAnchors, trustConfig })` .
- **Verification flags**: `createVerifySettings({ verifyTrust: true, verifyTimestampTrust: true })` — the signing certificate is validated against the Trust List, and a cert that is expired at validation time but was within validity at the trusted-timestamp time is accepted on that basis.
- **Trust configuration**: the EKU allow-list (`store.cfg`) constrains acceptable extended key usages to the C2PA-recognized set.
- A signing certificate chaining to a Trust List CA (e.g. `SSL.com C2PA ECC/RSA Root CA 2025` , the OriPics production CA) is reported `trusted` ; the current sandbox/dev certificate is self-signed and therefore reported `untrusted` (expected until production credentials are issued).

- In self-signed sandbox mode (`ORIPICS_C2PA_CA_PEM` set), the dev CA is used as the trust anchor instead; production mode (no `ORIPICS_C2PA_CA_PEM`) uses the C2PA Trust List above.

> **Note on SDK version**: OriPics pins `@contentauth/c2pa-node v0.5.x` (c2pa-rs v0.82.1). When this version validates an ingested third-party manifest whose per-image certificate is already expired (e.g. a Google Pixel Camera ingredient), it reports `signingCredential.expired` even though the manifest carries a trusted timestamp proving validity at signing time; conformant validators (e.g. current `c2patool`) resolve such a manifest to `trusted` . This affects only the *ingredient's* status; the OriPics active manifest is validated independently.

C.2.5. Protections Against Interception and/or Modification of Traffic

Encryption of Network Traffic

(Required for Assurance Level 1 for Distributed Implementation Class)

TLS versions and cryptographic protocols:

Communication Channel	Protocol	Notes
Edge (all) → Vercel Backend (API calls)	TLS 1.3	Enforced by Vercel CDN/Edge; HTTPS redirect enforced
Edge (all) → Supabase Storage (PNG upload via signed URL)	TLS 1.3	Supabase enforced
Backend → Supabase (DB + Storage)	TLS 1.3	Enforced via `tls.DEFAULT_MIN_VERSION = 'TLSv1.3'` in `src/instrumentation.ts` ; Supabase endpoints support TLS 1.3
Backend → SSL.com eSigner CSC API	TLS 1.3	SSL.com enforced
Backend → SSL.com TSA (timestamp.ssl.com)	TLS 1.3	SSL.com enforced

- **All network communication** between subsystems is encrypted with TLS 1.3
- Vercel enforces TLS 1.3 for all inbound connections; HTTP is automatically redirected to HTTPS
- No plaintext HTTP communication is used between any subsystems
- The `final_hash_hex` transmitted in the JWT chain is protected by TLS in transit and by the HMAC-SHA256 JWT signature at rest

C.2.6. Protections Against Exploitation of Hosting Environment

(Required for Assurance Level 1 for Distributed Implementation Class)

1. IAM system and security boundaries:

- **Vercel**: Team-based RBAC. The OriPics Vercel project is owned by a single team with one owner (Yongseog Son). MFA is required for the team owner account. Vercel enforces isolation between projects and between customers at the infrastructure level (SOC 2 Type II certified)
- **Supabase**: Row-Level Security (RLS) is enabled on all tables. The application uses `service_role` key for server-side operations (never exposed to Edge clients). Edge clients use `anon` key with RLS policies restricting access to

authenticated users' own data
- **GitHub**: Repository is private. Branch protection on `main` requires PR approval and CI passage. Force push is disabled
- **Edge clients**: No signing keys, no backend secrets, no `service_role` or `anon` keys embedded in Edge code. Edge clients authenticate users via NextAuth session cookies; all privileged operations require a valid session

2. Access policies for human and non-human principals:

Principal	Access	MFA
Vercel team owner (human)	Full project access, env var management	Yes
Vercel Functions runtime (non-human)	Read env vars, execute functions	N/A
Supabase service_role (non-human)	Full database + storage access	N/A (API key, server-only)
Supabase anon key (non-human)	RLS-restricted read/write	N/A (API key + user JWT)
GitHub repository admin (human)	Source code, CI/CD, secrets	Yes
Dependabot (non-human)	Create PRs for dependency updates	N/A (GitHub-managed)
Edge client / end user (non-human)	NextAuth session only; no infrastructure access	N/A

3. IAM policies for main cloud resources:

- **Vercel Functions**: Only deployed code from the `main` branch can execute in Production. Preview deployments run in isolated environments with separate env vars
- **Supabase Storage** (`oripics-proofs` bucket): Accessed only via service_role key from Backend Functions. Signed upload URLs for Edge clients have a 5-minute TTL and are path-restricted to the specific PNG being uploaded. No public write access; public read is restricted to published links only
- **Supabase PostgreSQL**: All tables have RLS enabled. `service_role` bypasses RLS for server-side operations; `anon` key is RLS-restricted

4. Vulnerability scanning and security review:

- GitHub Dependabot + CodeQL + Secret Scanning (as described in §C.2.3)
- OWASP Top 10 coverage via CodeQL rules
- Vercel and Supabase perform their own infrastructure-level security management as part of their respective SOC 2 Type II programs

5. Vulnerability remediation process:

- CRITICAL: Target fix within 30 days
- HIGH: Target fix within 90 days
- MODERATE: Target fix within 180 days
- LOW: Tracked and addressed on a best-effort basis
- Dependabot automatically creates PRs; team triages within 7 days
- Evidence of historical remediation: 21 advisories triaged 2026-05-09, Next.js security patch applied same day

C.2.7. Capture-Only Content Policy and Pre-Existing Manifest Handling

Content Policy and TOE Boundary

OriPics is a **photo provenance platform**. The `digitalCapture` DST is asserted only when the Generator Product can cryptographically confirm that the asset was captured by a hardware camera under the control of an authenticated device:

Client	Tier	TOE	Action	DST	Ingredient
OriPics Web	Standard	Outside TOE	`c2pa.opened`	None	uploaded file, `parentOf`, unknown provenance
OriPics iOS	Standard (file pick)	In TOE (device auth'd)	`c2pa.opened`	None	uploaded file, `parentOf`, unknown provenance
OriPics iOS	Verified (camera capture)	In TOE (App Attest)	`c2pa.created`	`digitalCapture`	None
OriPics Android	Standard (file pick)	In TOE (device auth'd)	`c2pa.opened`	None	uploaded file, `parentOf`, unknown provenance
OriPics Android	Verified (camera capture)	In TOE (Play Integrity)	`c2pa.created`	`digitalCapture`	None

`c2pa.created` (asserting creation of the asset) is issued only for Verified-tier captures, where device attestation (App Attest / Play Integrity) confirms in-app camera hardware capture; `digitalCapture` accompanies it. All file-pick / Web uploads are ingested via `c2pa.opened` because OriPics did not create the asset. OriPics does not certify AI-generated images, screenshots, or composites.

Pre-Existing C2PA Manifest Handling

When an uploaded image contains a pre-existing C2PA manifest (e.g., a photo previously signed by another C2PA generator product such as a Pixel Camera, Leica M11-P, or another conformant tool), OriPics detects this using `@contentauth/c2pa-node` `Reader.fromAsset()` before constructing the new manifest.

Implemented: Approach A — Ingredient (parentOf)

OriPics implements **Approach A**: every ingested file is recorded as a `c2pa.ingredient` assertion (`relationship: "parentOf"`) in the new OriPics manifest, under a `c2pa.opened` action. When a pre-existing C2PA manifest is detected, that manifest is preserved in the ingredient, maintaining the full provenance chain; when none is present, the ingredient simply records the uploaded file as unknown provenance.

Implementation (`apps/web/src/lib/oripics-stamp/c2pa.ts`):

1. Before constructing the new manifest, `Reader.fromAsset()` attempts to read the active manifest from the input PNG
2. **Verified tier** (mobile native camera, App Attest / Play Integrity confirmed): OriPics created the asset via attested hardware capture, so the primary action is **`c2pa.created`** with `digitalSourceType: digitalCapture`; no ingredient is added

3. **All ingested files** (Standard-tier Web/file-pick uploads, and any upload — verified or not — that already carries a prior C2PA manifest): OriPics did not create the asset, so the primary action is `c2pa.opened` (C2PA spec §9.3.2) and the ingested file is recorded as a `parentOf` ingredient via `builder.addIngredient(... relationship: "parentOf")` + `builder.setIntent('edit')`:
   - **Prior C2PA manifest present**: the ingredient preserves that manifest, maintaining the full provenance chain
   - **No prior manifest** (plain upload): the ingredient has **unknown provenance** — OriPics neither created it nor can prove its C2PA provenance

Scenario	Action	Ingredient
Verified tier, mobile camera, no prior C2PA	`c2pa.created` + `digitalCapture` DST	None
Standard tier, web/file-pick, no prior C2PA	`c2pa.opened` (no DST)	uploaded file, `parentOf`, **unknown provenance**
Any tier, prior C2PA manifest present	`c2pa.opened` (no DST)	`c2pa.ingredient.v3` with `relationship: "parentOf"` (prior manifest preserved)

(Note on Verified tier): Mobile native camera captures (Verified tier) produce fresh in-app captures with no prior C2PA manifest and are the only case where OriPics asserts `c2pa.created`. Every other path ingests an existing file via `c2pa.opened`.

Attachments

The following sample output files are provided with this submission:

1. **sample-oripics-standard.png** — Sample PNG with OriPics C2PA manifest (Standard tier, sandbox cert). Action: `c2pa.opened` (no `digitalSourceType`); the uploaded file is recorded as a `parentOf` ingredient of **unknown provenance**. `validation_state: Valid` (only the expected sandbox `signingCredential.untrusted` notice).
2. **sample-oripics-standard-manifest.json** — Extracted C2PA manifest JSON from the above sample.
3. **sample-oripics-with-ingredient.png** — Sample PNG demonstrating §C.2.7 ingestion of a prior-manifest file: a Google Pixel Camera image (`PXL_20250708_194721212.MP.jpg` from the C2PA Interoperability Testing Library) processed by OriPics. Action: `c2pa.opened`; the prior Pixel Camera manifest is preserved as `c2pa.ingredient.v3` with `relationship: "parentOf"`. The OriPics active manifest is valid (sandbox `untrusted` only); the additional `signingCredential.expired` / `untrusted` statuses belong to the **ingested Pixel ingredient**, whose per-image certificate is already expired and is validated through its trusted timestamp by conformant validators (see §C.2.4 §5 SDK note).
4. **sample-oripics-with-ingredient-manifest.json** — Extracted C2PA manifest JSON from the above sample.

(Note: Samples are signed with a sandbox test certificate, so the OriPics active manifest is reported `untrusted` (expected). The manifest structure is identical to production; when production credentials are issued from SSL.com — a CA on the C2PA Trust List — the active manifest is reported `trusted`. The validation logic (§C.2.4 §5) loads the C2PA Trust List in both cases.)